

1 Building a Bibliometric Shiny App

1.1 Learning objectives

After completing this chapter, you will be able to:

- Design a Shiny app architecture for bibliometric analysis
- Implement reactive data fetching from OpenAlex
- Build interactive UI components: dropdowns, sliders, and tabbed panels
- Create server-side logic for citation analysis and network visualisation
- Deploy a Shiny app to shinyapps.io or a self-hosted server

1.2 Setup

```
library(tidyverse)
library(shiny)
library(openalexR)
library(glue)

set.seed(20260509)

source(here::here("R", "sci_palette.R"))
```

1.3 Conceptual background

Shiny transforms R scripts into web applications. Users interact through widgets (dropdowns, sliders, buttons) and see results update in real time. For bibliometrics, Shiny apps serve two audiences: researchers who want to explore data without writing code, and decision-makers who need accessible analytics dashboards.

A Shiny app has two components: the **UI** (user interface, defining layout and inputs) and the **server** (the R logic that reacts to user input and generates output). **Reactive programming** connects inputs to outputs: when a user changes a dropdown, only the computations that depend on that input are re-executed.

For bibliometric apps, the typical architecture involves:

1. **Input panel:** Select a journal, institution, or search query; choose date range and sample size.
2. **Data fetching:** Reactively query OpenAlex based on user selections.
3. **Analysis tabs:** Trend charts, citation distributions, author rankings, and optionally network visualisation.
4. **Export:** Download buttons for data (CSV) and figures (PNG).

`biblioshiny()` from the `bibliometrix` package [Aria2017] provides a ready-made Shiny interface for many standard analyses. Building a custom app gives full control over the analysis pipeline and user experience.

Deployment options include shinyapps.io (hosted by Posit, free tier available), Posit Connect (enterprise), and self-hosted via Shiny Server (open source).

1.4 Worked example

1.4.1 App structure

We demonstrate the key components of a bibliometric Shiny app. The code below shows the UI and server logic separately for clarity.

1.4.2 UI definition

```
ui <- fluidPage(  
  titlePanel("Bibliometric Explorer"),  
  
  sidebarLayout(  
    sidebarPanel(  
      textInput("journal_id", "OpenAlex Source ID:",  
        value = "S148561398"),  
      sliderInput("year_range", "Publication Years:",  
        min = 2015, max = 2023, value = c(2020, 2023)),  
      numericInput("sample_size", "Sample Size:", value = 200,  
        min = 50, max = 1000),  
      actionButton("fetch", "Fetch Data", class = "btn-primary"),  
      hr(),  
      downloadButton("download_csv", "Download CSV")  
    ),  
  
    mainPanel(  
      tabsetPanel(  
        tabPanel("Trends",  
          plotOutput("trend_plot"),  
          tableOutput("summary_table")),  
        tabPanel("Citations",  
          plotOutput("cite_dist"),  
          plotOutput("cite_boxplot")),  
        tabPanel("Data",  
          DT::dataTableOutput("data_table"))  
      )  
    )  
  )  
)
```

1.4.3 Server logic

```
server <- function(input, output, session) {  
  
  works_data <- eventReactive(input$fetch, {  
    withProgress(message = "Fetching from OpenAlex...", {  
      oa_fetch(  
        entity = "works",  
        primary_location.source.id = input$journal_id,  
        from_publication_date = paste0(input$year_range[1], "-01-01"),  
        to_publication_date = paste0(input$year_range[2], "-12-31"),  
        type = "article",  
        options = list(sample = input$sample_size, seed = 42)  
      ) |>  
      transmute(  
        id, title = display_name,  
        year = year(publication_date),  
        cited_by_count, oa_status  
      )  
    })  
  })  
  
  output$trend_plot <- renderPlot({  
    works_data$plot_trend()  
  })  
  
  output$summary_table <- renderTable({  
    works_data$summary_table()  
  })  
  
  output$cite_dist <- renderPlot({  
    works_data$plot_cite_dist()  
  })  
  
  output$cite_boxplot <- renderPlot({  
    works_data$plot_cite_boxplot()  
  })  
  
  output$data_table <- renderDT({  
    works_data$data_table()  
  })  
  
  output$download_csv <- downloadHandler(  
    filename = function() {"Bibliometric Explorer.csv"},  
    content = function() {  
      works_data$download_csv()  
    }  
  )  
  
}
```

```

    )
  })
})

output$trend_plot <- renderPlot({
  req(works_data())
  works_data() |>
    count(year) |>
    ggplot(aes(x = factor(year), y = n)) +
    geom_col(fill = palette_sci(1)) +
    labs(x = "Year", y = "Publications") +
    theme_sci()
})

output$cite_dist <- renderPlot({
  req(works_data())
  ggplot(works_data(), aes(x = cited_by_count)) +
    geom_histogram(binwidth = 5, fill = palette_sci(1), colour = "white") +
    labs(x = "Citations", y = "Count") +
    theme_sci()
})

output$cite_boxplot <- renderPlot({
  req(works_data())
  works_data() |>
    filter(!is.na(oa_status)) |>
    ggplot(aes(x = oa_status, y = cited_by_count + 1)) +
    geom_boxplot(fill = palette_sci(1), alpha = 0.7) +
    scale_y_log10() +
    labs(x = "OA Status", y = "Citations (log)") +
    theme_sci()
})

output$summary_table <- renderTable({
  req(works_data())
  works_data() |>
    summarise(
      Articles = n(),
      `Mean Citations` = round(mean(cited_by_count), 1),
      `Median Citations` = median(cited_by_count),
      `OA Rate` = scales::percent(mean(oa_status != "closed", na.rm = TRUE))
    )
})

output$data_table <- DT::renderDataTable({
  req(works_data())
  works_data() |>
    select(title, year, cited_by_count, oa_status) |>
    arrange(desc(cited_by_count))
})

output$download_csv <- downloadHandler(
  filename = function() paste0("biblio_data_", Sys.Date(), ".csv"),

```

```

    content = function(file) write_csv(works_data(), file)
  )
}

```

1.4.4 Running the app

```
shinyApp(ui = ui, server = server)
```

1.4.5 biblioshiny as an alternative

```

# bibliometrix provides a ready-made Shiny interface:
# library(bibliometrix)
# biblioshiny()

```

1.5 Diagnostics and interpretation

- **Responsiveness:** API calls can take several seconds. Use `withProgress()` to show loading indicators.
- **Error handling:** Wrap API calls in `tryCatch()` to display user-friendly error messages instead of stack traces.
- **Memory:** Each user session loads data into R memory. For apps with many concurrent users, pre-aggregate data or use database backends.
- **Testing:** Use `shinytest2` to automate testing of Shiny apps.

1.6 Limitations and responsible use

1.7 Limitations and responsible use

- **Shiny apps are not analyses.** They enable exploration but do not substitute for rigorous, documented analytical pipelines. Use Shiny for discovery, not for producing final results.
- **API rate limits.** Live OpenAlex queries from a Shiny app can hit rate limits if many users fetch data simultaneously. Use caching or pre-fetched data for production apps.
- **Maintenance.** Shiny apps require ongoing maintenance: API changes, package updates, and server costs. Plan for long-term sustainability before deploying.
- **Interpretive guardrails.** Non-technical users may misinterpret metrics. Build in tooltips, descriptions, and warnings — especially around citation-based indicators [hicks2015; dora2012].

1.8 Common pitfalls

1.9 Common pitfalls

- **Fetching data on every input change.** Use `eventReactive()` with an action button instead of `reactive()` to prevent re-fetching on every slider adjustment.
- **Not validating inputs.** Check that the journal ID exists and the date range is valid before making API calls.
- **Monolithic server functions.** Break the server logic into modules for maintainability as the app grows.
- **Deploying without testing.** Test with realistic data volumes before deploying. An app that works with 100 records may fail with 10,000.

1.10 Exercises

1. **Add network tab.** Extend the app with a tab that builds and displays a co-authorship network using `visNetwork`.
2. **Caching.** Implement server-side caching so that repeated queries with the same parameters return instantly.
3. **Deployment.** Deploy the app to shinyapps.io. Test with different journal IDs and date ranges.
4. **biblioshiny comparison.** Run `biblioshiny()` and compare its features with your custom app. What does biblioshiny offer that your app does not?

1.11 Solutions

Solutions are provided in 1.11.

1.12 Further reading

- @aria2017 — `bibliometrix` and `biblioshiny()` for turnkey bibliometric exploration.
- @priem2022 — OpenAlex API for reactive data fetching.
- @hicks2015 — The Leiden Manifesto: designing interfaces that prevent metric misuse.
- @dora2012 — DORA: responsible assessment in interactive tools.

1.13 Session info

```
sessionInfo()
```

```
#> R version 4.4.1 (2024-06-14)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.4 LTS
#>
#> Matrix products: default
#> BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p-r0.3.26.so; LAPACK version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8      LC_NUMERIC=C          LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8    LC_MONETARY=C.UTF-8   LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8      LC_NAME=C             LC_ADDRESS=C
#> [10] LC_TELEPHONE=C        LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats      graphics  grDevices  utils      datasets  methods    base
#>
#> other attached packages:
#> [1] shiny_1.13.0      DT_0.34.0
#> [3] plotly_4.12.0     uwot_0.2.4
```

```

#> [5] Matrix_1.7-0 word2vec_0.4.1
#> [7] stm_1.3.8 topicmodels_0.2-17
#> [9] quanteda.textstats_0.97.2 visNetwork_2.1.4
#> [11] ggraph_2.2.2 tidygraph_1.3.1
#> [13] igraph_2.3.2 quanteda_4.4
#> [15] pdftools_3.9.0 arrow_24.0.0
#> [17] bibliometrix_5.4.0 RefManageR_1.4.0
#> [19] bib2df_1.1.2.0 rcrossref_1.2.1
#> [21] gt_1.3.0 tidytext_0.4.3
#> [23] glue_1.8.1 openalexR_3.0.1
#> [25] lubridate_1.9.5 forcats_1.0.1
#> [27] stringr_1.6.0 dplyr_1.2.1
#> [29] purrr_1.2.2 readr_2.2.0
#> [31] tidyr_1.3.2 tibble_3.3.1
#> [33] ggplot2_4.0.3 tidyverse_2.0.0
#> [35] bookdown_0.46 fs_2.1.0
#> [37] rmarkdown_2.31
#>
#> loaded via a namespace (and not attached):
#> [1] bibtex_0.5.2 RColorBrewer_1.1-3 jsonlite_2.0.0
#> [4] magrittr_2.0.5 modeltools_0.2-24 farver_2.1.2
#> [7] vctrs_0.7.3 memoise_2.0.1 askpass_1.2.1
#> [10] base64enc_0.1-6 tinytex_0.59 htmltools_0.5.9
#> [13] contentanalysis_1.0.0 curl_7.1.0 broom_1.0.12
#> [16] janeaustenr_1.0.0 cellranger_1.1.0 htmlwidgets_1.6.4
#> [19] tokenizers_0.3.0 plyr_1.8.9 httr2_1.2.2
#> [22] cachem_1.1.0 dimensionsR_0.0.3 mime_0.13
#> [25] lifecycle_1.0.5 pkgconfig_2.0.3 R6_2.6.1
#> [28] fastmap_1.2.0 digest_0.6.39 patchwork_1.3.2
#> [31] shinycssloaders_1.1.0 rprojroot_2.1.1 RSpectra_0.16-2
#> [34] SnowballC_0.7.1 labeling_0.4.3 urltools_1.7.3.1
#> [37] timechange_0.4.0 mgcv_1.9-1 polyclip_1.10-7
#> [40] httr_1.4.8 compiler_4.4.1 here_1.0.2
#> [43] bit64_4.8.0 withr_3.0.2 S7_0.2.2
#> [46] backports_1.5.1 viridis_0.6.5 ggforce_0.5.0
#> [49] MASS_7.3-60.2 rappdirs_0.3.4 bibliometrixData_0.3.0
#> [52] tools_4.4.1 otel_0.2.0 stopwords_2.3
#> [55] zip_2.3.3 httpuv_1.6.17 rentrez_1.2.4
#> [58] nlme_3.1-164 promises_1.5.0 grid_4.4.1
#> [61] stringdist_0.9.17 reshape2_1.4.5 generics_0.1.4
#> [64] gtable_0.3.6 tzdb_0.5.0 rscopus_0.9.0
#> [67] ca_0.71.1 data.table_1.18.4 hms_1.1.4
#> [70] xml2_1.5.2 utf8_1.2.6 ggrepel_0.9.8
#> [73] pillar_1.11.1 nsyllable_1.0.1 vroom_1.7.1
#> [76] later_1.4.8 splines_4.4.1 tweenr_2.0.3
#> [79] lattice_0.22-6 FNN_1.1.4.1 bit_4.6.0
#> [82] tidyselect_1.2.1 tm_0.7-18 miniUI_0.1.2
#> [85] knitr_1.51 gridExtra_2.3 NLP_0.3-2
#> [88] stats4_4.4.1 crul_1.6.0 xfun_0.57
#> [91] graphlayouts_1.2.3 matrixStats_1.5.0 humaniformat_0.6.0
#> [94] stringi_1.8.7 lazyeval_0.2.3 qpdf_1.4.1
#> [97] yaml_2.3.12 evaluate_1.0.5 codetools_0.2-20
#> [100] httpcode_0.3.0 cli_3.6.6 xtable_1.8-8
#> [103] dichromat_2.0-0.1 Rcpp_1.1.1-1.1 readxl_1.4.5

```

#> [106]	triebeard_0.4.1	XML_3.99-0.23	parallel_4.4.1
#> [109]	assertthat_0.2.1	pubmedR_1.0.2	slam_0.1-55
#> [112]	viridisLite_0.4.3	scales_1.4.0	crayon_1.5.3
#> [115]	openxlsx_4.2.8.1	rlang_1.2.0	fastmatch_1.1-8